

Python Code

```
1  import pygame
2  import sys
3  import os
4
5  def resource_path(relative_path):
6      try:
7          base_path = sys._MEIPASS
8      except Exception:
9          base_path = os.path.abspath(".")
10
11     return os.path.join(base_path, relative_path)
12
13 pygame.init()
14
15 screenW = 1440
16 screenH = 720
17 screen = pygame.display.set_mode((screenW, screenH))
18 clock = pygame.time.Clock()
19
20 floorY = screenH - 120
21
22 # Player Properties
23 playerW = 20
24 playerH = 20
25 playerX = 50
26 playerY = floorY
27 playerSpeed = 5
28 gravity = 1
29 jumpForce = 15
30
31 velY = 0
32 isJump = False
33
34
35 start_ticks = 0
36 timer_started = False
37 time_string = "00:00.00"
38 final_time_string = ""
39
40 fontPath = resource_path("roboto/Roboto-Regular.ttf")
41 font1 = pygame.font.Font(fontPath, 50)
42 font2 = pygame.font.Font(fontPath, 40)
43 font_timer = pygame.font.Font(fontPath, 35)
44
45 endGametxt = font1.render("Congrats", True, (0,0,0), (72,72,72))
46 startOvertxt = font2.render("Press 'R' to start over", True, (0,0,0), (72,72,72))
47 endGame = endGametxt.get_rect(center=(screenW/2, screenH/2))
48 startOver = startOvertxt.get_rect(center=(screenW/2, screenH/2+50))
49
50 # Game Platforms
```

```
51 platforms = [  
52     pygame.Rect(100, 570, 100, 50),  
53     pygame.Rect(300, 520, 100, 50),  
54     pygame.Rect(500, 470, 100, 50),  
55     pygame.Rect(700, 420, 100, 50),  
56     pygame.Rect(700, 320, 100, 50),  
57     pygame.Rect(700, 220, 100, 50),  
58     pygame.Rect(850, 400, 100, 50),  
59     pygame.Rect(850, 300, 100, 50),  
60     pygame.Rect(850, 200, 100, 50),  
61     pygame.Rect(950, 200, 25, 250),  
62     pygame.Rect(1050, 400, 100, 50),  
63     pygame.Rect(1250, 500, 290, 50)  
64 ]  
65 LevelIndex = 0  
66 running = True  
67 reachedEnd = False  
68  
69 while running:  
70     for ev in pygame.event.get():  
71         if ev.type == pygame.QUIT:  
72             running = False  
73             pygame.quit()  
74             sys.exit()  
75  
76     keys = pygame.key.get_pressed()  
77  
78     if not reachedEnd:  
79         if not timer_started:  
80             movement_keys = [  
81                 keys[pygame.K_a], keys[pygame.K_LEFT],  
82                 keys[pygame.K_d], keys[pygame.K_RIGHT],  
83                 keys[pygame.K_w], keys[pygame.K_UP], keys[pygame.K_SPACE]  
84             ]  
85             if any(movement_keys):  
86                 timer_started = True  
87                 start_ticks = pygame.time.get_ticks()  
88  
89             if (keys[pygame.K_a] or keys[pygame.K_LEFT]) and playerX > 0:  
90                 playerX -= playerSpeed  
91             if (keys[pygame.K_d] or keys[pygame.K_RIGHT]) and playerX < screenW - playerW:  
92                 playerX += playerSpeed  
93  
94             playerX_HB = pygame.Rect(playerX, playerY + 2, playerW, playerH - 4)  
95  
96             for plat in platforms:  
97                 if playerX_HB.colliderect(plat):  
98                     if keys[pygame.K_d] or keys[pygame.K_RIGHT]:  
99                         playerX = plat.left - playerW  
100                     if keys[pygame.K_a] or keys[pygame.K_LEFT]:  
101                         playerX = plat.right  
102  
103             if (keys[pygame.K_SPACE] or keys[pygame.K_w] or keys[pygame.K_UP]) and not  
isJump:
```

```
104         velY = -jumpForce
105         isJump = True
106
107     # Gravity Stuff
108     velY += gravity
109     playerY += velY
110
111     playerHB = pygame.Rect(playerX, playerY, playerW, playerH)
112
113     onPlatform = False
114
115     for plat in platforms:
116         if playerHB.colliderect(plat):
117             if velY > 0:
118                 if (playerHB.bottom - velY) <= plat.top:
119                     playerY = plat.top - playerH
120                     velY = 0
121                     isJump = False
122                     onPlatform = True
123             elif velY < 0:
124                 if (playerHB.top - velY) >= plat.bottom:
125                     playerY = plat.bottom
126                     velY = 0
127
128     if playerY >= floorY and playerX <= 200:
129         playerY = floorY
130         velY = 0
131         isJump = False
132         onPlatform = True
133
134     if not onPlatform and velY == 0:
135         isJump = True
136
137     if playerY > screenH:
138         playerX = 50
139         playerY = floorY
140         velY = 0
141         isJump = False
142         timer_started = False
143         time_string = "00:00.00"
144
145     if playerX >= screenW - playerW:
146         reachedEnd = True
147         final_time_string = time_string
148
149     else:
150         if keys[pygame.K_r]:
151             playerX = 50
152             playerY = floorY
153             velY = 0
154             isJump = False
155             reachedEnd = False
156             timer_started = False
157             time_string = "00:00.00"
```

```
158
159
160     if timer_started and not reachedEnd:
161         elapsed_ms = pygame.time.get_ticks() - start_ticks
162         minutes = elapsed_ms // 60000
163         seconds = (elapsed_ms % 60000) // 1000
164         milliseconds = (elapsed_ms % 1000) // 10
165         time_string = f"{minutes:02d}:{seconds:02d}.{milliseconds:02d}"
166     elif reachedEnd:
167         time_string = final_time_string
168
169
170     timer_color = (238, 75, 43) if reachedEnd else (0, 0, 0)
171     timer_surface = font_timer.render(time_string, True, timer_color)
172
173
174     screen.fill((182,182,182))
175
176     pygame.draw.rect(screen, (0,0,0), [0, 620, 200, 100])
177     for object in platforms:
178         pygame.draw.rect(screen, (0,0,0), object)
179
180     pygame.draw.rect(screen, (65,65,65), pygame.Rect(playerX, playerY, playerW,
181 playerH))
182
183     screen.blit(timer_surface, (30, 30))
184
185     if reachedEnd:
186         screen.blit(endGametxt, endGame)
187         screen.blit(startOvertxt, startOver)
188
189     pygame.display.flip()
190     clock.tick(60)
```